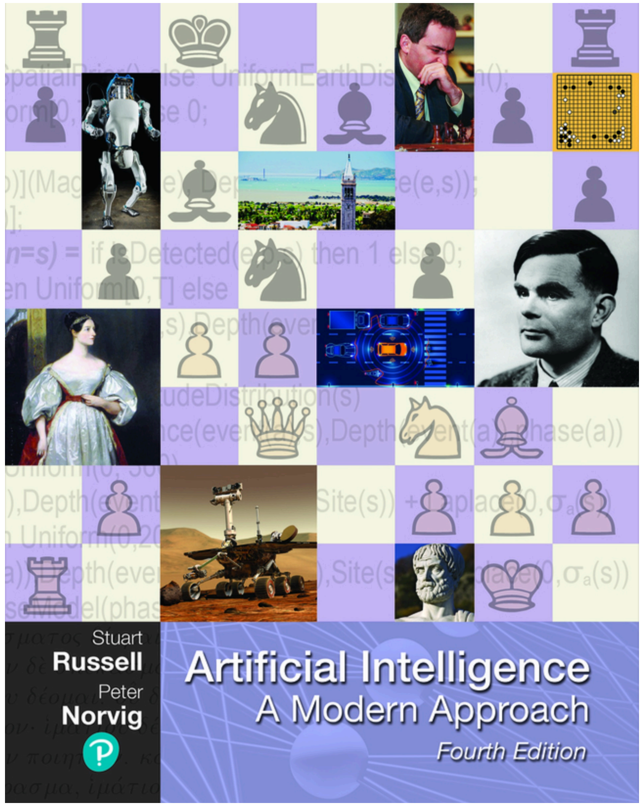
Artificial Intelligence and Machine Learning

Christian Cabrera Jojoa
Senior Research Associate and Affiliated Lecturer
Department of Computer Science and Technology
University of Cambridge
chc79@cam.ac.uk

The ML Context

Artificial Intelligence (AI)

"The field of **Artificial Intelligence** (AI) is concerned with *understanding and building intelligent entities.*"



Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach (4th Edition)

AI Dimensions

	Human	Rational
Thought		
Behaviour		

AI Dimensions

	Human	**Rational**
Thought	Systems that *think like humans*, focusing on cognitive and mental models that *mimic human reasoning*. • Example: Context-aware chatbots	Systems that *think rationally*, using logical principles and formal reasoning to *solve problems and make decisions*. • Example: Rule-based problem solvers
Behaviour	Systems that *act like humans*, exhibiting behaviours and interactions that are *natural and intuitive for human users*. • Example: Human-like robots	Systems that *act rationally*, making optimal decisions based on available *information and defined goals*. • Example: Optimal decision makers

AI Dimensions

	Human	Rational
Thought	• Cognitive modeling • Human-like reasoning • Natural language understanding • Common sense reasoning	• Logic-based systems • Theorem proving • Knowledge representation • Expert systems
Behaviour	• Natural language processing • Computer vision • Robotics • Human-computer interaction	• Optimization algorithms • Game theory • Planning systems • Decision theory

AI Foundations

AI Foundations

Philosophy Mathematics Economics Neuroscience

Psychology Computer Engineering Control Theory

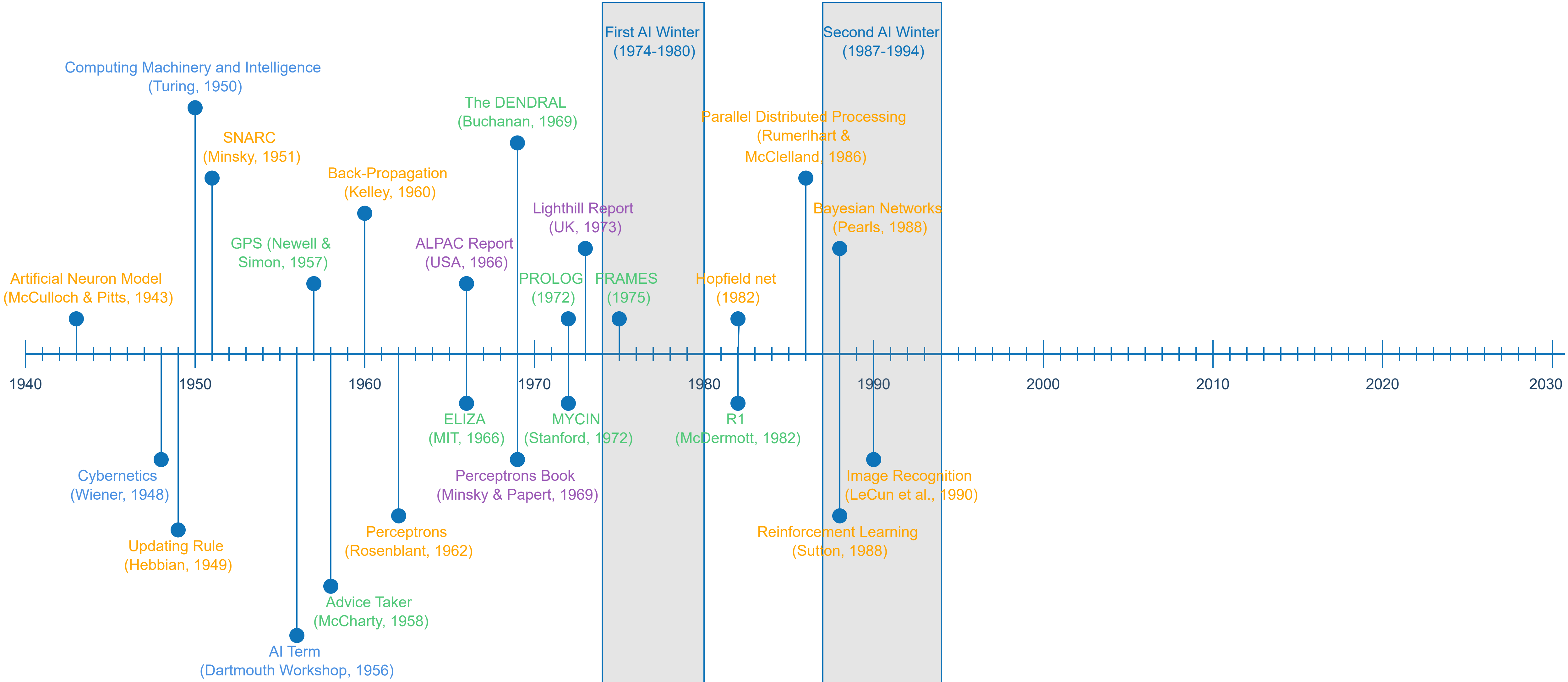
Linguistics Mind-body problem Logic Reasoning Ethics Probability

Statistics Optimization Graph theory Decision theory Game theory

Neural networks Learning Cognition Memory Algorithms Systems

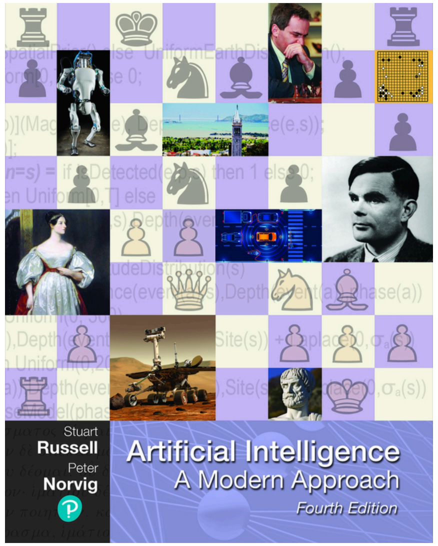
Feedback Language NLP Communication

AI History


First AI Winter (1974-1980)
Second AI Winter (1987-1994)
Computing Machinery and Intelligence (Turing, 1950)
SNARC (Minsky, 1951)
Back-Propagation (Kelley, 1960)
The DENDRAL (Buchanan, 1969)
Parallel Distributed Processing (Rumerlhart & McClelland, 1986)
Lighthill Report (UK, 1973)
Bayesian Networks (Pearls, 1988)
GPS (Newell & Simon, 1957)
ALPAC Report (USA, 1966)
Artificial Neuron Model (McCulloch & Pitts, 1943)
PROLOG (1972)
FRAMES (1975)
Hopfield net (1982)
1940
1950
1960
1970
1980
1990
2000
2010
2020
2030
ELIZA (MIT, 1966)
MYCIN (Stanford, 1972)
R1 (McDermott, 1982)
Cybernetics (Wiener, 1948)
Perceptrons Book (Minsky & Papert, 1969)
Image Recognition (LeCun et al., 1990)
Updating Rule (Hebbian, 1949)
Perceptrons (Rosenblant, 1962)
Reinforcement Learning (Sutton, 1988)
Advice Taker (McCharty, 1958)
AI Term (Dartmouth Workshop, 1956)

Artificial Intelligence - Intelligent Entities Rows

The field of **Artificial Intelligence** (AI) is concerned with *understanding and building intelligent entities.*



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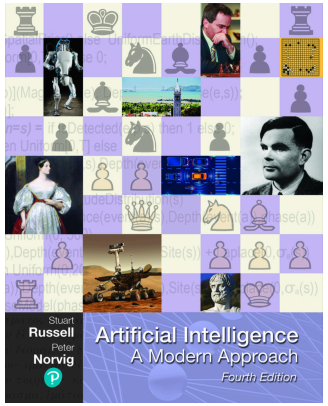
Artificial Intelligence - Intelligent Entities Rows and Columns



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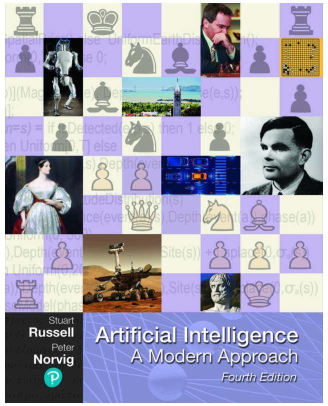

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Code Example

```
from sklearn.datasets import make_blobs
import matplotlib.pyplot as plt
# Generate sample data
X, y = make_blobs(n_samples=100, centers=2, random_state=42)
# Plot the data
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Sample ML Dataset")
plt.show()
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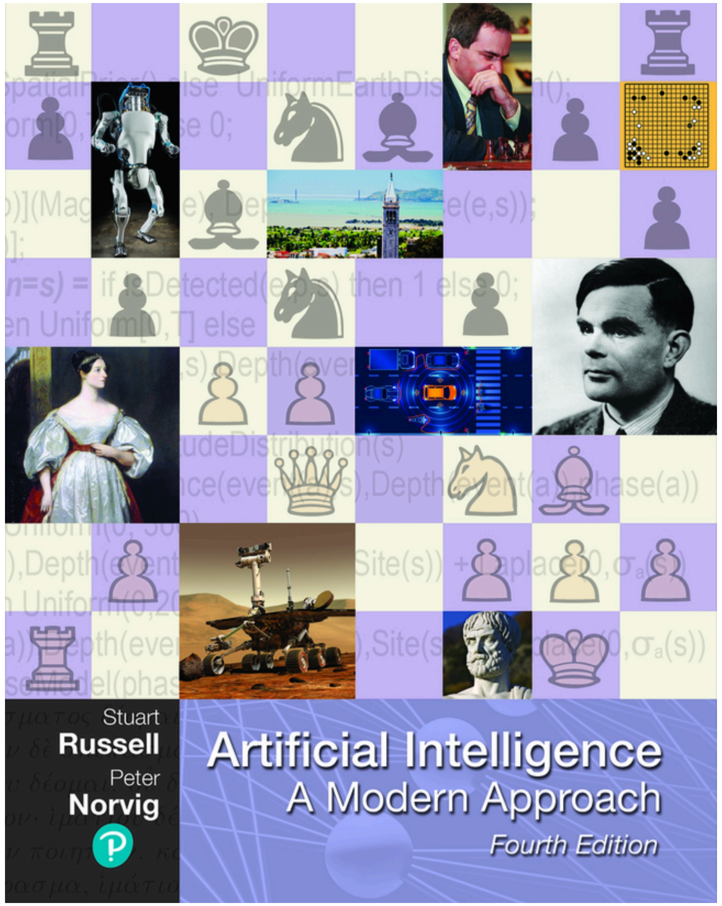
Code Example in Columns



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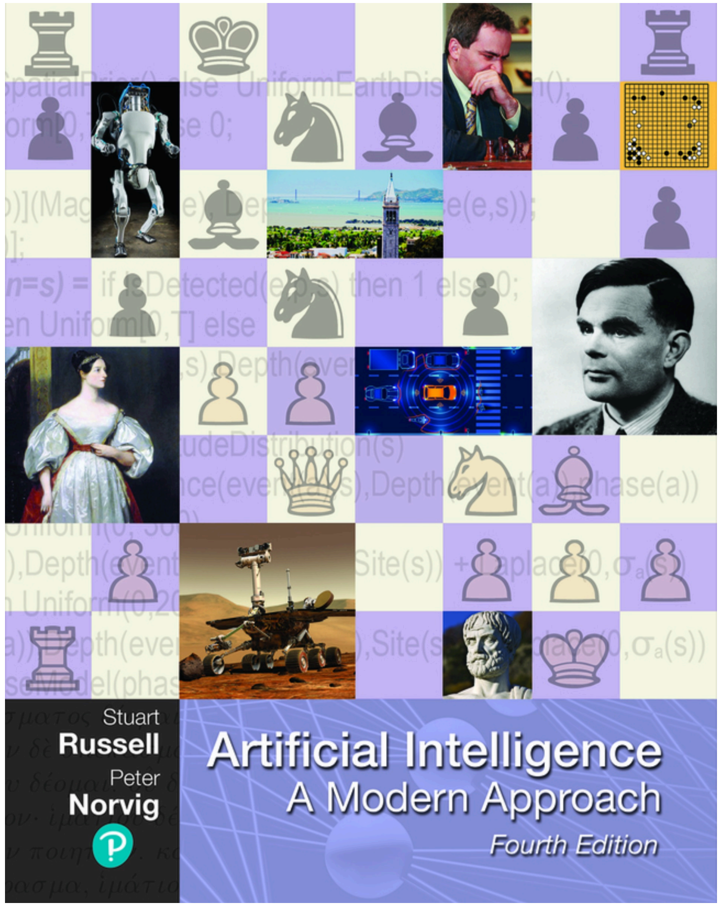

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History of Artificial Intelligence

The field of Artificial Intelligence has evolved through several key periods:

- 1943: McCulloch and Pitts proposed a model of artificial neurons - 1950: Alan Turing published "Computing Machinery and Intelligence" introducing the Turing Test - 1956: The Dartmouth Conference, where the term "Artificial Intelligence" was coined

The Early Years (1940s-1950s)

- Initial enthusiasm gave way to disappointment - Limited computing power and data availability
- Overpromising and underdelivering on AI capabilities

The First AI Winter (1974-1980)

- Rise of rule-based systems - Success in specific domains like medical diagnosis - Commercial applications of AI began to emerge

The Expert Systems Era (1980s)

- Expert systems proved expensive to maintain - Limited scalability of early AI approaches - Reduced funding and interest in AI research

The Second AI Winter (Late 1980s-1990s)

- Machine Learning revolution - Deep Learning breakthroughs - Big Data and increased computing power - Widespread practical applications - Integration into everyday technology

The Modern Era (2000s-Present)

- 1997: IBM's Deep Blue defeats chess champion Garry Kasparov - 2011: IBM's Watson wins Jeopardy! - 2012: AlexNet revolutionizes computer vision - 2016: AlphaGo defeats world champion Lee Sedol - 2020s: Large Language Models and Generative AI

This historical context helps us understand how AI has evolved from theoretical concepts to practical applications that impact our daily lives.

Key Milestones

Public Perception of AI

The public's understanding and perception of Artificial Intelligence has evolved significantly over time:

- AI as human-like robots (Hollywood effect) - AI having consciousness or emotions - AI being either completely benevolent or malevolent - AI being a single, unified technology

Common Misconceptions

- Science fiction shaping public expectations - Sensationalist headlines about AI capabilities - Fear-mongering about job displacement - Overemphasis on existential risks

Media Influence

- AI systems are specialized tools, not general intelligence - Current AI lacks consciousness or self-awareness - AI systems reflect their training data and human biases - AI is a collection of different technologies and approaches

Reality vs. Fiction

- Job displacement and economic impact - Privacy and surveillance - Bias and discrimination - Decision-making transparency - Security and misuse

Public Concerns

- Potential for medical breakthroughs - Environmental monitoring and protection - Educational assistance - Accessibility improvements - Scientific research acceleration

Understanding these perceptions is crucial for:

- Setting realistic expectations
- Addressing legitimate concerns
- Promoting responsible AI development
- Fostering public trust in AI technologies

Positive Perceptions

The ML Concept

What is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence that focuses on developing systems that can learn from and make decisions based on data.

- A system's ability to automatically learn and improve from experience - Without being explicitly programmed - Through pattern recognition and statistical inference

Core Definition

Key Components

1. **Data** - Training data - Validation data - Test data - Features and labels

2. **Model**
    - Mathematical representation
    - Learning algorithm
    - Parameters and weights

3. **Learning Process**
    - Pattern recognition
    - Parameter optimization
    - Performance evaluation

Types of Machine Learning

1. **Supervised Learning** - Learning from labeled data - Classification and regression - Examples: spam detection, price prediction

2. **Unsupervised Learning**
    - Finding patterns in unlabeled data
    - Clustering and dimensionality reduction
    - Examples: customer segmentation, anomaly detection
3. **Reinforcement Learning**
    - Learning through trial and error
    - Reward-based learning
    - Examples: game playing, robotics

- Feature engineering - Model training - Overfitting and underfitting - Bias-variance tradeoff - Generalization - Evaluation metrics

Fundamental Concepts

- Handling complex patterns - Adapting to new data - Automating decision-making - Extracting insights from large datasets - Solving problems where traditional programming is impractical

Why Machine Learning?

History of Machine Learning

Machine Learning has its roots in several disciplines and has evolved through distinct phases:

- 1943: McCulloch and Pitts' artificial neuron model - 1950: Alan Turing's learning machine concept - 1952: Arthur Samuel's checkers-playing program

Early Foundations (1940s-1950s)

- 1957: Frank Rosenblatt's Perceptron - Early neural network models - Initial optimism followed by limitations

The Perceptron Era (1950s-1960s)

- Reduced interest in ML research - Focus on expert systems - Limited computational resources

The AI Winter Impact (1970s-1980s)

- Backpropagation algorithm - Support Vector Machines - Decision Trees and Random Forests - Increased computational power

The Renaissance (1980s-1990s)

- Big Data revolution - Deep Learning resurgence - Cloud computing enabling large-scale ML - Open-source ML frameworks

The Modern Era (2000s-Present)

- 1995: Support Vector Machines - 2006: Deep Learning renaissance - 2012: AlexNet and ImageNet success - 2014: Generative Adversarial Networks - 2017: Transformer architecture - 2020s: Large Language Models

Key Breakthroughs

- Transfer Learning - Few-shot Learning - Self-supervised Learning - Federated Learning - Explainable AI - Ethical AI considerations

This historical perspective shows how ML has grown from theoretical concepts to practical tools that power modern technology.

Current Trends

Perception of Machine Learning

Machine Learning is perceived differently across various sectors and communities:

- Competitive advantage - Cost reduction - Process automation - Data-driven decision making - Innovation enabler

Industry Perspective

- Research frontier - Interdisciplinary field - Theoretical foundations - Experimental validation - Peer-reviewed progress

Academic View

- Often conflated with AI - Seen as "magic" or "black box" - Concerns about job displacement - Privacy considerations - Trust and transparency issues

Public Understanding

- Strategic investment - Digital transformation - ROI considerations - Talent acquisition - Risk management

Business Leaders

- Tool for problem-solving - Engineering challenges - Best practices - Open-source collaboration - Continuous learning

Technical Community

- Bias and fairness - Accountability - Transparency - Privacy protection - Social impact

Ethical Considerations

- ML can solve any problem - More data always means better results - ML models are objective - ML replaces human judgment - ML is only for large companies

Common Misconceptions

- ML as a tool, not a solution - Requires quality data - Needs domain expertise - Involves trial and error - Demands maintenance

Understanding these different perceptions helps in:

- Setting appropriate expectations
- Communicating effectively

Realistic Expectations

- Building trust
- Managing implementation
- Addressing concerns

The Objective Definition of ML

Subjective Aspects of Machine Learning

Machine Learning involves several subjective elements that influence its development and application:

- Domain expertise - Intuition in feature selection - Judgment in model evaluation - Interpretation of results - Decision-making thresholds

Human Factors

- Problem framing - Data collection methods - Feature engineering - Model selection - Hyperparameter tuning

Design Choices

- Value judgments in data selection - Fairness definitions - Privacy boundaries - Transparency requirements - Accountability standards

Ethical Considerations

- Regional data availability - Language and cultural context - Social norms and values - Regulatory environments - Market expectations

Cultural Influences

- Confirmation bias - Selection bias - Interpretation bias - Anchoring effects - Overconfidence

Personal Biases

- Success metrics selection - Performance thresholds - Error tolerance levels - Trade-off decisions
- Quality assessment

Subjectivity in Evaluation

- Research priorities - Funding decisions - Team composition - Project timelines - Resource allocation

Impact on Development

- Documentation - Peer review - Diverse perspectives - Continuous feedback - Regular audits

Understanding these subjective elements is crucial for:

- Building better ML systems
- Ensuring fairness
- Maintaining transparency
- Making informed decisions
- Building trust

Managing Subjectivity

Objective Aspects of Machine Learning

Machine Learning is grounded in several objective, measurable components:

- Linear algebra - Probability theory - Statistics - Calculus - Optimization

Mathematical Foundations

- Algorithms - Data structures - Computational complexity - Memory requirements - Processing speed

Technical Components

- Accuracy - Precision - Recall - F1 score - ROC curves - AUC scores

Performance Metrics

- Dataset size - Feature dimensions - Data types - Missing values - Outliers

Data Characteristics

- Processing power - Memory capacity - Storage requirements - Network bandwidth - Energy consumption

Computational Resources

- Number of parameters - Model architecture - Training time - Inference speed - Memory footprint

Model Properties

- Error rates - Loss functions - Convergence speed - Generalization ability - Robustness

Quality Measures

- Data preprocessing - Model training - Validation procedures - Testing protocols - Deployment pipelines

Standardized Processes

Objective Evaluation

- Reproducibility - Benchmarking - A/B testing - Statistical significance - Performance baselines

Understanding these objective aspects is essential for:

- Building reliable systems
- Making fair comparisons
- Ensuring reproducibility
- Optimizing performance
- Scaling solutions

ML Promises

Promises of Machine Learning

Machine Learning offers numerous potential benefits across various domains:

- Early disease detection - Personalized medicine - Drug discovery - Medical image analysis - Treatment optimization

Healthcare

- Process automation - Predictive maintenance - Supply chain optimization - Customer behavior analysis - Fraud detection

Business and Industry

- Data analysis acceleration - Pattern discovery - Hypothesis generation - Simulation and modeling - Cross-disciplinary insights

Scientific Research

- Climate modeling - Resource management - Pollution monitoring - Wildlife conservation - Disaster prediction

Environmental Protection

- Personalized learning - Adaptive testing - Educational content creation - Student performance analysis - Learning path optimization

Education

- Accessibility improvements - Language translation - Disaster response - Public health monitoring - Social service optimization

Social Impact

- Productivity gains - Cost reduction - New business models - Market efficiency - Job creation in new sectors

Economic Benefits

- Autonomous systems - Natural language processing - Computer vision - Robotics - Smart infrastructure

Technological Advancement

- Personal assistants - Smart homes - Transportation optimization - Healthcare accessibility - Information accessibility

Quality of Life

- Accelerated innovation - Cross-domain applications - New methodologies - Collaborative research - Knowledge discovery

These promises must be balanced with:

- Ethical considerations
- Privacy concerns

Research and Development

- Implementation challenges
- Resource requirements
- Social impact

ML Risks

Risks and Challenges in Machine Learning

Machine Learning implementation carries several significant risks that must be carefully managed:

- Model failure - Data quality issues - System vulnerabilities - Performance degradation - Integration challenges

Technical Risks

- Bias and discrimination - Privacy violations - Lack of transparency - Accountability gaps - Unintended consequences

Ethical Risks

- Job displacement - Digital divide - Social manipulation - Cultural impact - Dependency creation

Social Risks

- Adversarial attacks - Data breaches - Model theft - System manipulation - Privacy violations

Security Risks

- Implementation costs - ROI uncertainty - Market disruption - Resource allocation - Competitive pressure

Economic Risks

- Regulatory compliance - Liability issues - Intellectual property - Data protection - Contractual obligations

Legal Risks

- System reliability - Maintenance requirements - Scalability issues - Technical debt - Knowledge gaps

Operational Risks

- Energy consumption - Resource utilization - E-waste generation - Carbon footprint - Sustainability impact

Environmental Risks

- Public perception - Trust erosion - Brand damage - Stakeholder confidence - Market position

Reputational Risks

- Robust testing - Regular audits - Ethical guidelines - Security protocols - Continuous monitoring

Addressing these risks requires:

- Comprehensive planning
- Stakeholder engagement
- Risk assessment

Mitigation Strategies

- Contingency planning
- Ongoing evaluation

ML in Practice

Machine Learning in Practice

Implementing Machine Learning requires careful consideration of practical aspects:

Project Lifecycle

1. **Problem Definition** - Business understanding - Success criteria - Resource assessment - Timeline planning - Stakeholder alignment

2. **Data Preparation**
   - Data collection
   - Cleaning and preprocessing
   - Feature engineering
   - Data validation
   - Version control

3. **Model Development**
   - Algorithm selection
   - Model training

- Version control - Documentation - Testing - Monitoring - Continuous integration

Best Practices

- Python ecosystem - TensorFlow/PyTorch - Scikit-learn - Cloud platforms - MLOps tools

Tools and Frameworks

- Cross-functional teams - Clear communication - Knowledge sharing - Code review - Documentation standards

Team Collaboration

- Model validation - Performance testing - Security testing - Compliance checking - User acceptance testing

Quality Assurance

- Model retraining - Performance monitoring - Bug fixing - Feature updates - Documentation updates

Maintenance

- Infrastructure needs - Cost management - Performance optimization - Resource allocation - Load balancing

Scaling Considerations

- Clear objectives - Quality data - Skilled team - Proper tools - Stakeholder support

Success Factors

- Data quality - Model drift - Resource constraints - Integration issues - Maintenance overhead

Common Challenges

- Technical expertise - Project management - Domain knowledge - Resource allocation - Continuous improvement

Implementing ML successfully requires:

Many Thanks!

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